

On Configuring, Calibrating, and Using the NanoVNA

This procedure is for configuring, calibrating, and using the inexpensive and popular NanoVNA to measure the SWR of a common Ham radio antenna.

There are several different hardware versions of the NanoVNA – The one used here is called the “NanoVNA-H”, hardware version “3.6_MS”. It has a 2.8 inch touchscreen display. The firmware is version 1.2.27 as collected from GitHub.

The NanoVNA kit came with most of what is needed to make SWR measurements. I picked up a set of SMA adapters off Amazon to allow me to connect the BNC or SO-239 from the end of the antenna feed line to the SMA connector on the NanoVNA.

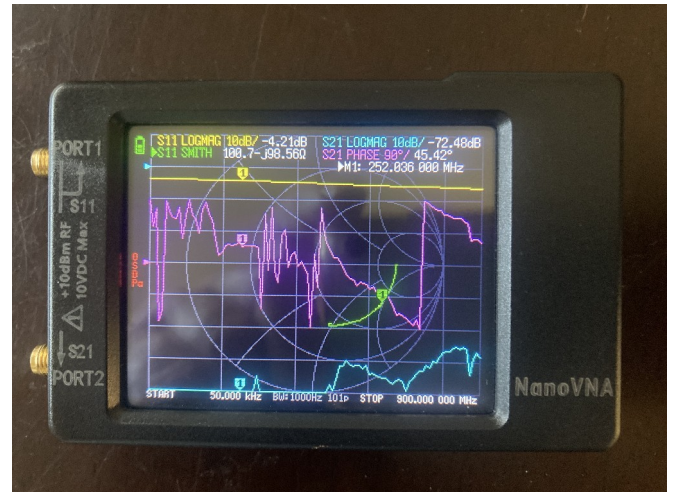
Reading the characteristics of an antenna only requires one of the two inputs on the NanoVNA – PORT1/S11/CH0 is used (upper connection). The other connector is not used in this procedure.

Initial Display Setup

Turn on the NanoVNA
(Your initial screen may look different.)

Here we see LOGMAG of S11, LOGMAG of S21, the ever exciting SMITH chart of S11, and PHASE of S21.

Note the scan range is all the way from 50 kHz to 900 MHz



Custom Configuration

The next steps will show you how to make the necessary trace changes and save a custom configuration.

Click anywhere on screen with a stylus to open the Main Menu.
Select 'DISPLAY'.
Select 'TRACE' – By default, 4 traces will be listed initially.

Click on a trace label to “select” it.
Click again un-highlight it and turn it off.
Turn off 'TRACE 2' and 'TRACE 3' as they are unnecessary.



Configure Traces

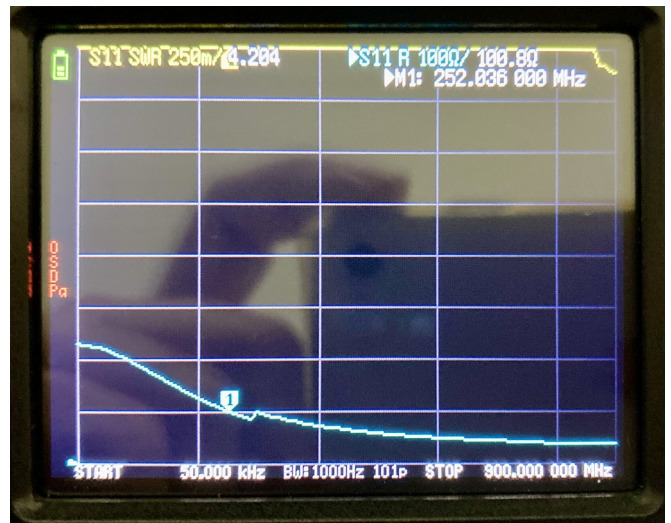
With 'TRACE 0' enabled (with color) and selected (checkbox), select 'BACK'.

Select 'FORMAT S11(REFL)', select 'SWR' (checkbox), and select 'BACK'.

Select 'TRACE' again, select 'TRACE 1' (checkbox), select 'BACK', select 'FORMAT S11(REFL)', select 'RESISTANCE', and select 'BACK' and 'BACK'.

We now see SWR of S11 and Resistance of S11.

The scan range (bottom line) is still 50 kHz to 900 MHz.



Set Desired Scan Range:

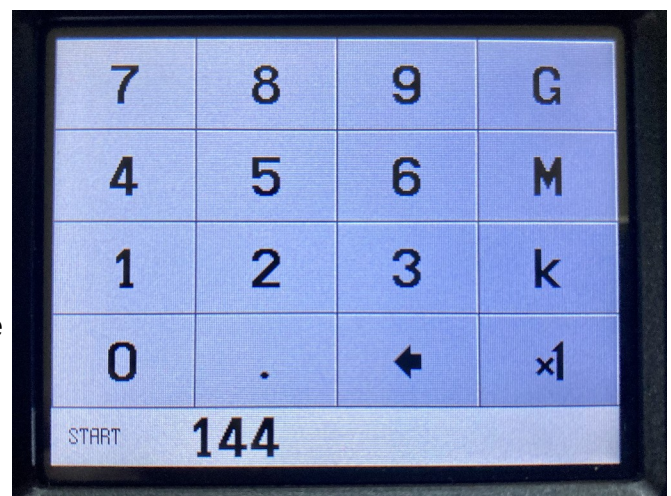
Select 'STIMULUS' from the main menu to set the Band Range (MHz)

Select 'START' and enter the lower-end of the frequency range (ex: "144 and M")

You must press either "G", "M" (Megahertz) or "k" (Kilohertz) after the appropriate number.

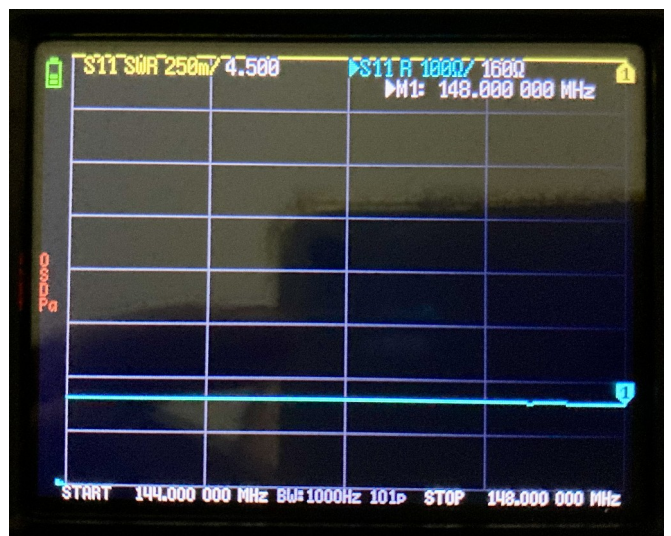
Click on 'STOP' and enter the high-end of the frequency range (ex: "148 and M")

Select 'BACK' to previous menu



Here we see the final configuration. Trace 0 (yellow) on Port 0 (S11) is showing SWR. Trace 1 (cyan) on Port 0 (S11) is showing resistance in Ohms.

The scan range is 144 MHz to 148 MHz.



Calibrate for the desired frequency range

Select 'CALIBRATE' to go to the Calibration Menu
Select 'RESET' (very important!) in order to delete the previous calibration constants and replace with specific calibration values needed for your desired frequency range.

Select 'CALIBRATE' to bring up the calibration menu

There will be a blue progress bar across the top line of the screen for each of these steps:

Screw on the OPEN calibration standard and Select "OPEN".

Screw on the SHORT calibration standard and Select "SHORT"

Screw on the LOAD calibration standard and Select "LOAD"

Select 'DONE' and you have the option of saving the Stimulus Range and Calibration in non-volatile memory locations S1-S4 (Not required, but a good idea)

TIP: Do NOT use Save 0 because factory calibration data is stored there.



Ready to scan your antenna

Here are NanoVNA screenshots of two scans of the same 2 meter $\frac{1}{4}$ wave ground plane antenna with two different feed lines, ...because RF.

